Mohammad Vazir Panah

HPC & Performance Software Engineer · C++, SYCL, Compilers

Fisciano (SA), Italy · (+39) 389 664 6511 · mohammad.vazirpanah@gmail.com

LinkedIn · GitHub · Website

Summary

Software engineer focused on performance on parallel and heterogeneous hardware, with an M.Sc. in Computer Systems Architecture and three years of R&D in the HPC group at ISISLab, University of Salerno. Experienced in C/C++, SYCL, OpenMP, OpenCL and CUDA, and in compiler-level work with GCC and LLVM on RISC-V vector processors. Used to systematic benchmarking, performance analysis and reproducible workflows (Git, Docker, CI, CMake).

Technical Skills

Programming	C, C++ (templates, modern C++), Python, Assembly
Parallel Programming	SYCL (AdaptiveCPP), OpenMP (including SIMD), OpenCL, CUDA, SIMD intrinsics
Compilers	GCC, LLVM/Clang, auto-vectorization analysis, RISC-V Vector extension (RVV 0.7 and 1.0)
Architectures	RISC-V, multicore CPUs, embedded GPUs (Imagination PowerVR), FPGA
Performance Work	Benchmarking, performance portability, performance analysis on real hardware
Simulation & Modeling	Gem5, QEMU, SystemC
Tools	Git, Docker, CI, CMake, Linux
Certification	TinyML, Harvard University (edX)

Professional Experience

Research Fellow, HPC Group (ISISLab) *2023 - 2026*

University of Salerno, Department of Computer Science, Salerno, Italy

- Co-developed an open-source RISC-V benchmark suite (151 TSVC loops and 7 real-world applications, RVV 0.7 and 1.0) and benchmarked GCC, LLVM and LLVM-EPI auto-vectorization on two RISC-V vector boards (AllWinner D1, BananaPi-F3); showed that LMUL tuning gives up to 3× speedup (IEEE PDP 2025)
- Worked on performance-portable SYCL programming (AdaptiveCPP) across embedded CPUs and GPUs in the LibreRT project; co-author of the system paper (ACM Computing Frontiers 2026)
- Debugged kernel dispatch failures on Imagination PowerVR GPUs down to the LLVM-based AdaptiveCPP runtime and fixed the kernel lookup logic; resolved related OpenCL build problems
- Implemented and evaluated applications with SYCL, OpenMP and OpenCL on heterogeneous CPU/GPU platforms
- Set up RISC-V toolchains and software stacks on physical boards for the EuroHPC LIGATE project
- Built reproducible benchmarking pipelines with Git, Docker, CI and CMake

Teaching Assistant *2020 - 2021*

Shahid Beheshti University, Tehran, Iran

- Advanced Computer Architecture, System-on-Chip, Design & Analysis of Real-Time Embedded Systems
- Ran student projects from environment setup to lab sessions, feedback and grading

C++ Instructor *2013 - 2015*

Sepehr Educational Institutions, Kashmar, Iran

- Taught C++ programming courses

Selected Projects

RISC-V Vector Benchmark Suite *2024 - 2025*

C, GCC, LLVM, GitHub

- Benchmark suite covering many data-level-parallelism patterns, used to compare compiler versions and find where auto-vectorization fails

LibreRT, Portable Real-Time Programming for Heterogeneous Platforms *2023 - 2026*

C++, SYCL, OpenMP, OpenCL, Project page

- Portable real-time programming stack for platforms from microcontrollers to edge devices with embedded GPUs

Parallel Sorting and Matrix Multiplication Architectures *2020*
Verilog and CUDA, GitHub

- Designed mesh- and hypercube-connected sorting networks in hardware, with a CUDA implementation of the hypercube version
- Designed a shared-memory matrix multiplication architecture on a multistage interconnection network

SADF Model of Computation in ForSyDe-SystemC (M.Sc. Thesis) *2022*
C++ and SystemC, GitHub

- Implemented the Scenario-Aware Dataflow model with C++ template metaprogramming; the core process constructors were merged into the official ForSyDe-SystemC repository

Real-Time Tiny YOLO Inference on FPGA *2019*
Intel DE10-Nano-SoC, GitHub

- Fixed-point CNN inference pipeline on FPGA under tight resource limits

Education

M.Sc. Computer Engineering, Computer Systems Architecture *2019 - 2022*
Shahid Beheshti University, Tehran, Iran

- GPA 18.43/20 · Courses include Advanced Computer Architecture, Parallel Systems Architecture, Interconnection Networks

B.Sc. Computer Engineering, Hardware Engineering *2012 - 2016*
Birjand University, Birjand, Iran

- GPA 16/20 · Top 15% of Computer Engineering graduates

Publications

- L. Carpentieri, **M. Vazir Panah**, B. Cosenza. *A Performance Analysis of Autovectorization on RVV RISC-V Boards.* IEEE PDP 2025.
- B. Cosenza et al. (incl. **M. Vazir Panah**). *LibreRT: Portable Heterogeneous Real-Time Programming for the Embedded Computing Continuum.* ACM Computing Frontiers 2026.
- S.-H. Attarzadeh-Niaki, **M. Vazir Panah**. *System-Level Modeling of Dynamic Applications with Scenario-Aware Dataflow Graphs.* JICSE 2023.
- **M. Vazir Panah**, S.-H. Attarzadeh-Niaki, A. Salimi-Badr. *ROS-Based Co-Simulation for Formal Cyber-Physical Robotic System Design.* IEEE CSICC 2022.

Awards

- National R&D Research Grant, Mobile Telecommunication Company of Iran (2020), awarded in competition with Master's and PhD students

Languages

Persian (native) · English (C1) · Italian (B1 listening, A2 speaking and writing)

References available on request.